

Multiobjective Optimization

Elitist Multi-Objective Evolutionary Algorithms

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- ➊ **Introduction to Elitism in MOEAs**
 - Why elitism is important
 - Challenges in multi-objective elitism
- ➋ **Rudolph's Elitist MOEA**
- ➌ **Elitist Non-Dominated Sorting GA (NSGA-II)**
- ➍ **Distance-Based Pareto GA (DPGA)**
- ➎ **Strength Pareto EA (SPEA)**
- ➏ **Pareto-Archived Evolution Strategy (PAES)**
- ➐ **Multi-Objective Messy GA (MOMGA)**
- ➑ **Other Elitist MOEAs & Summary**

Why Elitism Matters

In Single-Objective Optimization:

- Best solution(s) preserved across generations
- Monotonic improvement guaranteed (Rudolph, 1994)

Challenges in Multi-Objective Elitism:

1 Which solutions are "elite"?

- All non-dominated solutions are equally good
- Number of elites varies each generation

2 How much elitism is appropriate?

- Too little: lose good solutions
- Too much: lose diversity (premature convergence)

Common Elitism Strategies:

- Combine parent + offspring, select best
- Maintain external archive of elites
- Controlled elitism (adaptive)

Key Idea: Systematic elitism with convergence proof

Three-Phase Algorithm:

- ① **Phase 1:** Find non-dominated solutions of offspring Q_t
 - $Q^* =$ non-dominated solutions of Q_t
- ② **Phase 2:** Compare Q^* with parent population P_t
 - If $q \in Q^*$ dominates parent solutions: add to P'
 - If $q \in Q^*$ is non-dominated with all parents: add to Q'
- ③ **Phase 3:** Fill new population P_{t+1} in priority order:
 - ① Modified parents P_t
 - ② P' (elites that dominated parents)
 - ③ Q' (non-dominated with parents)
 - ④ Remaining offspring

Hand Calculation (Min-Ex)

Parent Population (P_t)			Offspring Population (Q_t)		
Sol.	f_1	f_2	Sol.	f_1	f_2
1	0.31	6.10	a	0.21	5.90
2	0.43	6.79	b	0.79	3.97
3	0.22	7.09	c	0.51	6.51
4	0.59	7.85	d	0.27	6.93
5	0.66	3.65	e	0.58	4.52
6	0.83	4.23	f	0.24	8.54

Phase 1: $Q^* = \{a, b, e\}$ (non-dominated in Q_t)

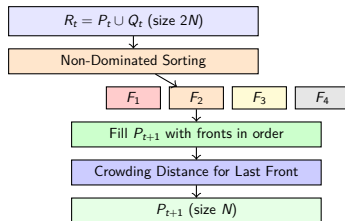
Phase 2:

- a dominates parents 1, 2, 3, 4 $\Rightarrow P' = \{a\}$
- b dominates parent 6 $\Rightarrow P' = \{a, b\}$
- e non-dominated with parent 5 $\Rightarrow Q' = \{e\}$

Phase 3: New population $P_{t+1} = \{5, a, b, c, d, e\}$

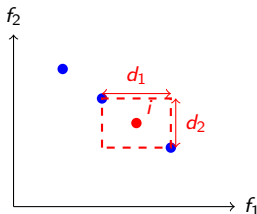
Key Innovations:

- 1 Elitism via combined parent+offspring population
- 2 Crowding distance for diversity (no σ_{share} parameter)
- 3 Fast non-dominated sorting ($O(MN^2)$)



Crowding Distance

Concept: Estimate density around a solution



Crowding Distance Calculation:

$$d_i = \sum_{m=1}^M \frac{f_m^{(i+1)} - f_m^{(i-1)}}{f_m^{\max} - f_m^{\min}}$$

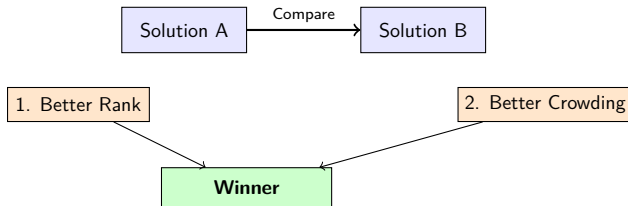
Properties:

- Boundary solutions get ∞ distance
- Larger distance = less crowded area
- Complexity: $O(MN \log N)$

Crowded Tournament Selection

Comparison Operator: Solution i wins if:

- 1 $r_i < r_j$ (better non-domination rank), OR
- 2 $r_i = r_j$ and $d_i > d_j$ (larger crowding distance)

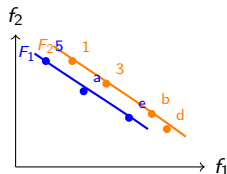


Advantages:

- No σ_{share} parameter needed
- Fast computation and works for any number of objectives

Hand Calculation (Min-Ex)

Combined Population $R_t = P_t \cup Q_t$



Resulting Selection:

- Front $F_1 = \{5, a, e\}$ selected completely.
- Front $F_2 = \{1, 3, b, d\}$ sorted by crowding distance.
- Select best 3 from F_2 to reach $N = 6$.
- **Final** P_{t+1} : $\{5, a, e, 3, b, 1\}$

Osyczka and Kundu (1995)

Key Idea: Fitness calculated based on distance from elite set.

Elite Set: Maintains all non-dominated solutions found so far.

Distance Formula:

$$d^{(k)}(x) = \sqrt{\sum_{m=1}^M (f_m(x) - e_m^{(k)})^2}, \quad d_{min}(x) = \min_k d^{(k)}(x)$$

Two Fitness Cases:

① **If x is dominated by an elite solution:**

$$F(x) = F(e_{min}) - d_{min}(x)$$

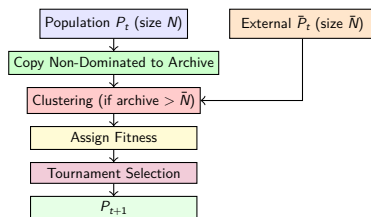
② **If x is non-dominated with elites:**

$$F(x) = F(e_{min}) + d_{min}(x)$$

(Add x to elite set, remove dominated elites)

Disadvantage: Elite set size can grow without bound.

Key Idea: Maintain an external archive of non-dominated solutions.



Fitness Assignment

Step 1: Strength of External Solutions ($i \in \bar{P}_t$)

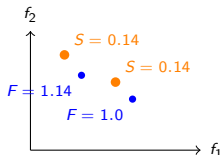
$$S_i = \frac{|\{j \in P_t : i \succeq j\}|}{N+1}$$

- $S_i \in [0, 1)$. Higher strength = dominates more solutions.

Step 2: Fitness of Population Members ($j \in P_t$)

$$F_j = 1 + \sum_{i \in \bar{P}_t, i \succeq j} S_i$$

- Smaller fitness = better.
- External solutions always have better fitness ($F < 1$).

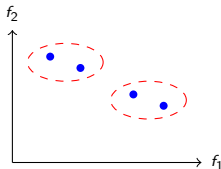


Clustering Algorithm

Goal: Reduce external population size to maximum limit $\bar{N}$.

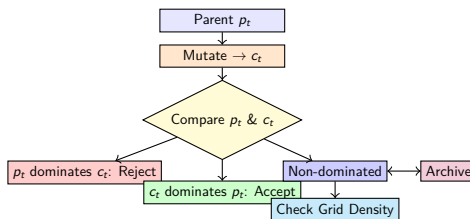
Procedure:

- 1 Start: each solution represents a separate cluster.
- 2 Compute cluster distances (average pair-wise distance).
- 3 Merge two closest clusters iteratively.
- 4 Repeat until exactly $\bar{N}$ clusters remain.
- 5 Keep the solution closest to each cluster centroid.



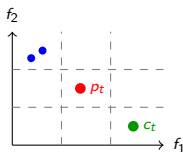
Note: Always retain extreme solutions in separate clusters to preserve spread.

Key Idea: $(1 + 1)$ -ES local search using an archive for diversity.



Density Check with Hypercubes

Key Idea: Divide objective space grid into adaptive hypercubes.



Decision Criteria (when p_t and c_t are non-dominated):

- If c_t resides in a less crowded hypercube $\Rightarrow$ Accept c_t as new parent.
- Otherwise $\Rightarrow$ Retain p_t .

Parameters:

- Archive size $\bar{N}$
- Depth d (Number of grid divisions = $(2^d)^M$)

Messy GA Background:

- Identifies and combines optimal building blocks.
- Uses variable-length chromosomes.

MOMGA Modifications:

① Multiple Templates:

- One locally optimal template per objective.
- Fills missing bits using appropriate objective templates.

② Niche Tournament Selection:

- Emphasizes non-dominated solutions.

③ External Archive:

- Stores non-dominated solutions found during search.

① NSAGA (Annealing GA):

- Combines non-dominated sorting with Simulated Annealing.
- Convergence proof via temperature reduction.

② PCGA (Pareto Converging GA):

- Uses rank histogram for convergence termination.
- $O(MN^3)$ complexity; no explicit niching operator.

③ Micro-GA (μ GA):

- Uses very small population size (4–5 individuals).
- Excellent for computationally expensive evaluations.

④ ERMOCS:

- Uses coevolutionary shared niching (CSN).

Key Features Comparison

Algorithm	Elitism Strategy	Diversity Mechanism	Complexity
Rudolph's	Parent+Offspring priority	None	$O(MN^2)$
NSGA-II	Combined population	Crowding distance	$O(MN^2)$
SPEA	External archive	Clustering	$O(MN^2)$
PAES	Archive + hypercubes	Hypercube density	$O(MN \log N)$
DPGA	Elite set	Distance to elites	$O(MN^2)$
MOMGA	External archive	Messy GA + niching	$O(MN^2)$

Observations:

- **NSGA-II** and **SPEA** are the most widely adopted standards.
- **NSGA-II** avoids setting arbitrary niche parameters (σ_{share}).

① Elitism is Crucial:

- Improves convergence to Pareto front and guarantees monotonicity.

② NSGA-II (Recommended Standard):

- Fast non-dominated sorting and parameter-free crowding distance.

③ SPEA / SPEA2:

- Highly effective external archive approach using clustering.

④ PAES:

- Simple $(1 + 1)$ -ES local search, ideal for quick implementations.

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